

PERSONAL

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INTERESTS

Solving the many problems that occur in a biological system requires an understanding of how the system works. To understand a system, we must understand, at multiple levels of analysis, the principles by which it operates. How can we design careful experiments and tools to unveil these principles? A good attempt at answering this question does not rely on a singular approach, but creates meaningful connections between them.

EDUCATION

Carleton College, Northfield, MN
 Bachelor of Arts in Cognitive Science (expected 2028)
 Neuroscience minor. Mathematics minor. 3.89 GPA.

**RESEARCH
EXPERIENCE**

Research Assistant, PI: **Dr. Joel A. Tripp**
 Neuroscience Program, Carleton College, Northfield, MN
 September 2025 - Present

- Investigating the neural circuitry of the vocal-motor system in Alston's singing mice to determine how neurobiological substrates mediate socially relevant behavior.
- Conducting neuropeptide anatomy experiments to map modulatory influences on vocal-motor circuits, advancing understanding of the neurochemical basis of social behaviors.
- Analyzing and visualizing spectrogram vocalization data

Research Assistant, PI: **Dr. David J. Herzfeld**, Co-Mentor: **Dr. Aaron Suminski**

University of Wisconsin School of Medicine & Public Health, Madison, WI

November 2025 - Present (May-Aug 2026 funded by *National Science Foundation* and *College of Engineering*)

- Designing and conducting novel behavioral experiments in rhesus monkeys to dissect the cerebellar basis of motor memories.
- Designed a converter for visualization of Full Binary Pursuit spike-sorting outputs in Phy, resolving a long-standing integration barrier in the lab's electrophysiology pipeline.
- Revealed a time-delay property of action potentials that allows for exhaustive detection in extracellular recordings, reducing the sensitivity-specificity tradeoff that limits reproducibility in systems neuroscience.

PUBLICATIONS

Jones† & Herzfeld. Exploiting deterministic dynamics of action potentials for precise and exhaustive spike sorting: a new approach. In Preparation.

Szycher A.D.†, **Jones, M.O.†**, Janik J.R., and **Joel A. Tripp**. Distribution of vasopressin and oxytocin neurons in Alston's singing mouse. In Preparation.

Bond, R.†, Brown, L., **Jones, M.O.**, and **Joel A. Tripp**. Ghrelin suppresses courtship behavior in male mice. In Preparation.

**CONFERENCE
PRESENTATIONS**

Bond, R.†, Brown, L., **Jones, M.O.**, and **Joel A. Tripp**. Hunger Hormone Ghrelin Suppresses Courtship Behavior in Male Mice. Soc. Behav. Neuroendocrinology, Niagara Falls, NY. 2026. Presented.

Szycher A.D., **Jones, M.O.**, Janik J.R., and **Joel A. Tripp**. Characterizing the distribution of vasopressin and oxytocin neurons and their relation to vocal-motor circuitry in Alston's singing mouse. Midbrains Conf., Sioux Falls, SD. 2025. Presented.

Szycher A.D., **Jones, M.O.**, Janik J.R., and **Joel A. Tripp**. Characterizing the distribution of vasopressin and oxytocin neurons and their relation to vocal-motor circuitry in Alston's singing mouse. Soc. Behav. Neuroendocrinology, Online. 2025. Attended.

**RELEVANT
COURSEWORK**

Neuro/Bio: Foundations in Neuroscience (w Lab), Neurons Circuits & Behavior (w Lab)

Seminar: Topics in Neuroscience, Genes Evolution & Development (w Lab)

Math/CS: Linear Algebra, Intro to Computer Science, Data Structures, Psychological Statistics

**TECHNICAL
SKILLS**

Programming: Python, Kotlin, R, Julia

Experimental Techniques: Confocal Laser Microscopy, Multi-Contact NHP Neurophysiology, Spike Sorting, Behavioral Data Collection, Rodent Brain Cryosectioning, Fluorescent Immunohistochemistry

Tools: Spikeinterface, Matplotlib, UnitRefine, Bombcell, NumPy, SciPy, Phy, Kilosort, MountainSort, SpyKing Circus, Full Binary Pursuit, DeepSqueak

**HONORS &
AWARDS**

2026, NSF SURE-REU Fellow, UW-Madison College of Engineering.

2026, NCAA DIII All-Conference Honors, Track & Field.

2025, Towsley Conference Travel Award, Carleton College.

2024, Academic All-State Honors, Football.

2024, DI All-Conference Honors, Baseball.

2023, DI All-State Honors, School Record, Swimming.

**TEACHING
EXPERIENCE**

Sophomore Spring, *Hormones, Brain, and Behavior*

Carleton College, Teaching Assistant & Grader

LANGUAGES

English (native), German (proficient)